

Question

Yellow solid **A** is composed of two elements, a metallic element **X** and a non-metallic element **Y**, and is one of the important minerals of element **X**. Calcination of **A** in air can produce solid **B** and colorless gas **C**. **B** contains 69.94% of element **X**. **C** reacts with Cl_2 to obtain a colorless fuming liquid **D**. **D** is easily hydrolyzed to produce **E** and **F**. **C** can react with O_2 to produce **G** under the action of a heated catalyst. **G** is soluble in water and also yields **E**.

C is an excellent Lewis acid. Researchers recently discovered that the reaction of 0.5 mL of anhydrous **C** with 0.02 g of a certain metal azide **I** at low temperature can produce an adduct **H**. The nitrogen content in **H** is 13.87%. If the orientation of **C** in **H** is ignored, its anionic symmetry is the same as that of water.

The following statements are correct:

- A.** All other options are incorrect
- B.** **A** is a 1:2 type ionic compound.
- C.** The highest oxidation state that element **X** can form is +4.
- D.** **D** is a regular tetrahedral molecule
- E.** **G** has a certain reducing property.
- F.** The mass fraction of nitrogen in **I** is approximately 28%
- G.** The mass fraction of **H** cation is 43.9%
- H.** **H** is a 2:1 type ionic compound.

Answer

Correct Answer: **G**

Detailed Explanation

B is obtained by calcining **A**, it can be determined that **B** is the stable oxide of element **X**; let the chemical formula of **B** be XO_n , containing 69.94% of element **X**.

CHECKPOINT

1 PTS

B is the stable oxide of element **X**

If it is XO , the relative atomic mass of **X** is $16/(1 - 0.6994) - 16 = 37.2$, no reasonable metal.

If it is X_2O_3 , the relative atomic mass of **X** is $(16 \times 3/(1 - 0.6994) - 16 \times 3)/2 = 55.9$, which is Fe.

If it is XO_2 , the relative atomic mass of **X** is $16 \times 2/(1 - 0.6994) - 16 \times 2 = 74.5$, no reasonable metal.

Other oxide types also have no reasonable results; therefore **X** is Fe, **B** is Fe_2O_3 . Fe exists in the +6 valence state, option C is incorrect.

CHECKPOINT

2 PTS

X is Fe, exists in the +6 valence state

CHECKPOINT

1 PTS

B is Fe_2O_3

Minerals generally contain sulfur or carbon elements; gas **C** may be SO_2 or CO_2 ; but **C** can react with Cl_2 , so CO_2 is excluded; **C** is likely to be SO_2 .

CHECKPOINT

1 PTS

Minerals generally contain sulfur or carbon elements

CHECKPOINT

1 PTS

C is likely to be SO_2

C reacts with O_2 to produce **G**, then **G** is SO_3 ; dissolving in water to obtain **E**, then **E** is H_2SO_4 . SO_3 is the highest valence oxide and has no reducing property, option E is incorrect.

CHECKPOINT

1 PTS

G is SO_3 , has no reducing property

At this time, **Y** should be the sulfur element, so **A** should be pyrite FeS_2 . The FeS_2 anion is S_2^{2-} , which is a 1:1 type ionic compound, option B is incorrect.

CHECKPOINT

2 PTS

Y should be the sulfur element, so **A** should be pyrite FeS_2

CHECKPOINT

1 PTS

FeS_2 anion is S_2^{2-} , which is a 1:1 type ionic compound

SO_2 reacts with Cl_2 to obtain a colorless fuming liquid, according to the elemental properties, it is sulfuryl chloride, that is, **D** is SO_2Cl_2 , which hydrolyzes to produce sulfuric acid **E** and hydrochloric acid **F**. SO_2Cl_2 is not a regular tetrahedron, but a distorted tetrahedral molecule, option D is incorrect.

CHECKPOINT

1 PTS

D is SO_2Cl_2 , which is a distorted tetrahedral molecule

I is a metal azide, which reacts with sulfur dioxide to obtain an adduct, so there should be no redox reaction; the chemical formula of the adduct can be written as $\text{M}(\text{SO}_2)_x(\text{N}_3)_y$, considering that the anion symmetry is the same as water, x and y should be 1 and 2 respectively;

CHECKPOINT

1 PTS

Reacts with sulfur dioxide to obtain an adduct, so there should be no redox reaction

CHECKPOINT

1 PTS

Anion symmetry is the same as water, the ratio of SO_2, N_3 is 1:2 or 2:1

When x is 1 and y is 2, the atomic mass of metal M is $14 \times 6 / 0.1387 - 14 \times 6 - 64 = 457.6$, the atomic mass is too large.

x is 2; at this time, the atomic mass of metal M is $14 \times 3 / 0.1387 - 14 \times 3 - 64 \times 2 = 132.8$, which is Cs , just in line with the valence.

Therefore, the chemical formula of **H** is $\text{Cs}[(\text{SO}_2)_2\text{N}_3]$, and **I** is CsN_3 . $\text{Cs}[(\text{SO}_2)_2\text{N}_3]$ is a 1:1 type ionic compound, option H is incorrect.

CHECKPOINT

2 PTS

The chemical formula of **H** is $\text{Cs}[(\text{SO}_2)_2\text{N}_3]$, which is a 1:1 type ionic compound

CHECKPOINT

1 PTS

I is CsN_3

Calculation shows that option F is incorrect and G is correct.